BINARY

UNIT 2.1

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HOW TO CONVERT A DECIMAL NUMBER INTO BINARY (POWERS OF 2)

- To achieve this will utilize the **powers of 2.**

2 ^ 0 = **1**

2 ^ 1 = **2**

2 ^ 2 = **4**

2 ^ 3 = **8**

2 ^ 4 = **16**

2 ^ 5 = **32**

2 ^ 6 = **64**

2 ^ 7 = **128**

```
>>> bin(75)
'0b1001011'
>>>
```

- For example, let's convert **75** into binary

128 64 32 16 8 4 2 1

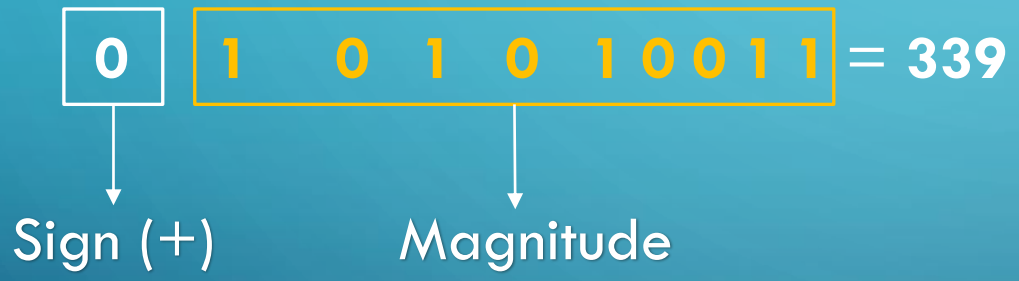
0 **1 0 0 1 0 1 1** = **75**

Sign (+) Magnitude

HOW TO CONVERT A DECIMAL NUMBER INTO BINARY (POWERS OF 2) (CONTINUES…)

- Let's take another example of **339**

512 256 128 64 32 16 8 4 2 1



- **1 = True**
- **0 = False**

```
>>> int(True)
1
>>> int(False)
0
>>>
```

```
>>> bin(339)
'0b101010011'
>>>
```

HOW TO CONVERT A DECIMAL NUMBER INTO BINARY SUCCESSIVE DIVISION

- Now, let's use **successive division** to find the binary of *75*

75 / 2 = 37 R **1**

37 / 2 = 18 R **1**

18 / 2 = 9 R **0**

9 / 2 = 4 R **1**

4 / 2 = 2 R **0**

2 / 2 = 1 R **0**

1 / 2 = 0 R **1**

- We now get the binary from the bottom to the top, making it:

1001011

```
>>> bin(75)
'0b1001011'
>>> 0b1001011
75
>>>
```

HOW TO ADD BINARY NUMBERS

- Here are some basics of addition:

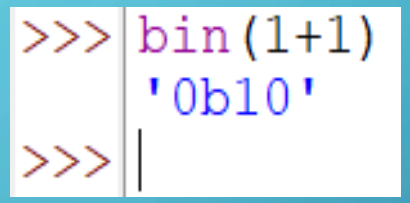
0 + 0 = **0**

0 + 1 = **1**

1 + 0 = **1**

1 + 1 = **10**

```
   1
+  1
____
```



1 **0**, this is the same as 1 + 1 = 2

2 / 2 = **1** R **0**

HOW TO ADD BINARY NUMBERS (CONTINUES…)

```
    1
    1
+   1
-----
  1 1
```

```
>>> bin(1+1+1)
'0b11'
>>>
```

This is the same as 1+1+1 = 3

3 / 2 = **1** R **1**

- **NOTE:** On the bottom we are putting the **reminder** (**R**) and **carrying** the number of times 2 enters into the target number.

HOW TO ADD BINARY NUMBERS (CONTINUES…)

- Let's take an example below:

```
  1 0 1 0  = 10
+ 1 0 0 1  =  9
-----------  ---
  1 0 0 1 1 = 19
```

16	8	4	2	1	Decimal
	1	0	1	0	10
	1	0	0	1	9
1	0	0	1	1	19

```
>>> bin(10)
'0b1010'
>>> bin(9)
'0b1001'
>>> bin(19)
'0b10011'
>>>
```

HOW TO SUBTRACT BINARY NUMBERS

- Here are some basics of subtraction:

1 - 0 = 1

1 - 1 = 0

```
>>> bin(8-1)
'0b111'
>>>
```

	1	1	
0	~~2~~	~~2~~	2
~~1~~	~~0~~	~~0~~	0
-			1
0	1	1	1

→

8	4	2	1
8			
-			1
			7

1 0 → 0 2

- 1 → 1

1 In base 2, will get a 2 from the nearest value.

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1'S & 2'S COMPLEMENT

1'S COMPLEMENT

- Obtain 1's complement of **1010**.
- **NOTE:** In ones complement we **slip** the numbers, 1 becomes 0 and 0 becomes 1.
- Therefore 1's complement of **1010** is **0101**.

2'S COMPLEMENT

- Formular:

  2's complement = 1's complement + 1

QUESTION: Obtain 1's complement of **10111010**.

1s complement = **01000101**

2s complement = **01000110**

SIGNED MAGNITUDE

CONVERTING A BINARY

- Let's take an example, let's express (**-6**) using the signed magnitude.

1. What is (**+6**) in binary?

8 4 2 1

0 1 1 0 = +6

Sign (+) ← 0 | 1 1 0 → Magnitude

NEGATIVE BINARY REPRESENTATION

2. The **Sign (+)** will change from **0** to **1** **Sign (-).**

8 4 2 1

Sign (-) ← **1 1 1 0**

NOTE: 1 represent **negative** and **0** **positive**

2'S COMPLEMENT

NEGATIVE NUMBERS

- Represent **-12** in binary using the 2's complement.

16 8 4 2 1

0 1 1 0 0 = **12**

1 0 0 1 1 = 1's complement

2'S COMPLEMENT (-12)

1 0 0 1 1

\+ 1

---

1 0 1 0 0 = **-16** + **4** = **-12**

Sign (-) Magnitude

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2'S COMPLEMENT SUBTRACTION

BINARY CONVERSATION

- Let's subtract **39 – (25).**
- Using normal math operation 39 – 25 = 14.

1. Convert 39 and 25 into binary.

64 32 16 8 4 2 1

0 1 2 2

0 1 0 0 1 1 1 = **39**

- 0 0 1 1 0 0 1 = **25**

0 0 0 1 1 1 0 = **14**, Proof (**8+4+12**)

1'S & 2'S COMPLEMENT

2. Find the 1's complement 25.

1 1 0 0 1 1 0 = **25** 1's complement

3. Find the 2's complement of 25.

1 1 0 0 1 1 1 = **25** 2's complement

NOTE: We are targeting **25,** because in 2's complement subtraction we **change the sign** of the number after the minus (-) sign, so **25** becomes **-25.**

2'S COMPLEMENT SUBTRACTION (CONTINUES…)

2'S COMPLEMENT SUBTRACTION

4. Finally, do the 2's complement subtraction of 39 and 25.

  0 1 0 0 1 1 1

+1 1 0 0 1 1 1

(1) 0 0 0 1 1 1 0

NOTE: We have a carry 1 at the end, but we are just ignoring it, meaning an **overflow** as occurred.

VERIFICATION

- Is 0001110 14?

  128 64 32 16 8 4 2 1

  0 0 0 0 1 1 1 0

- 8 + 4 + 2 = 14, **NOTE:** In 2's complement subtraction we **add** two binaries, think of it as 39 – 25 is the same as 39 + (-25).

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PERT CHART

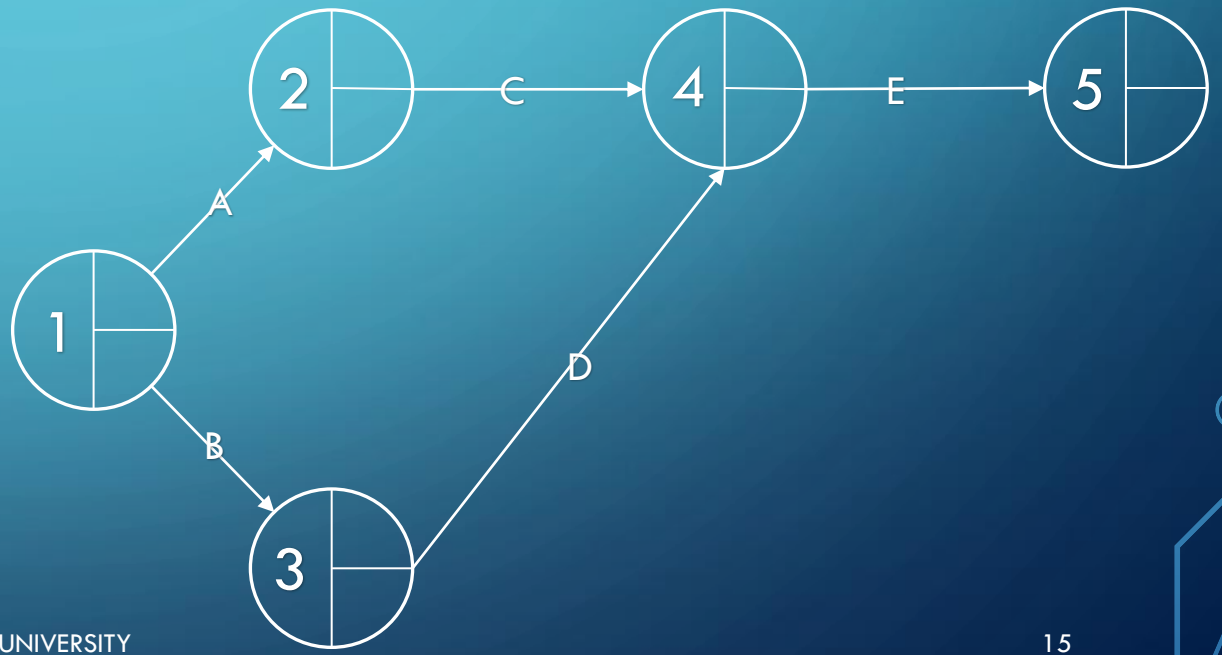
UNIT 2.2

PERT CHART EXAMPLE 1

- Question

Activities	Dependencies
A	
B	
C	A
D	B
E	C,D

- PERT Chart

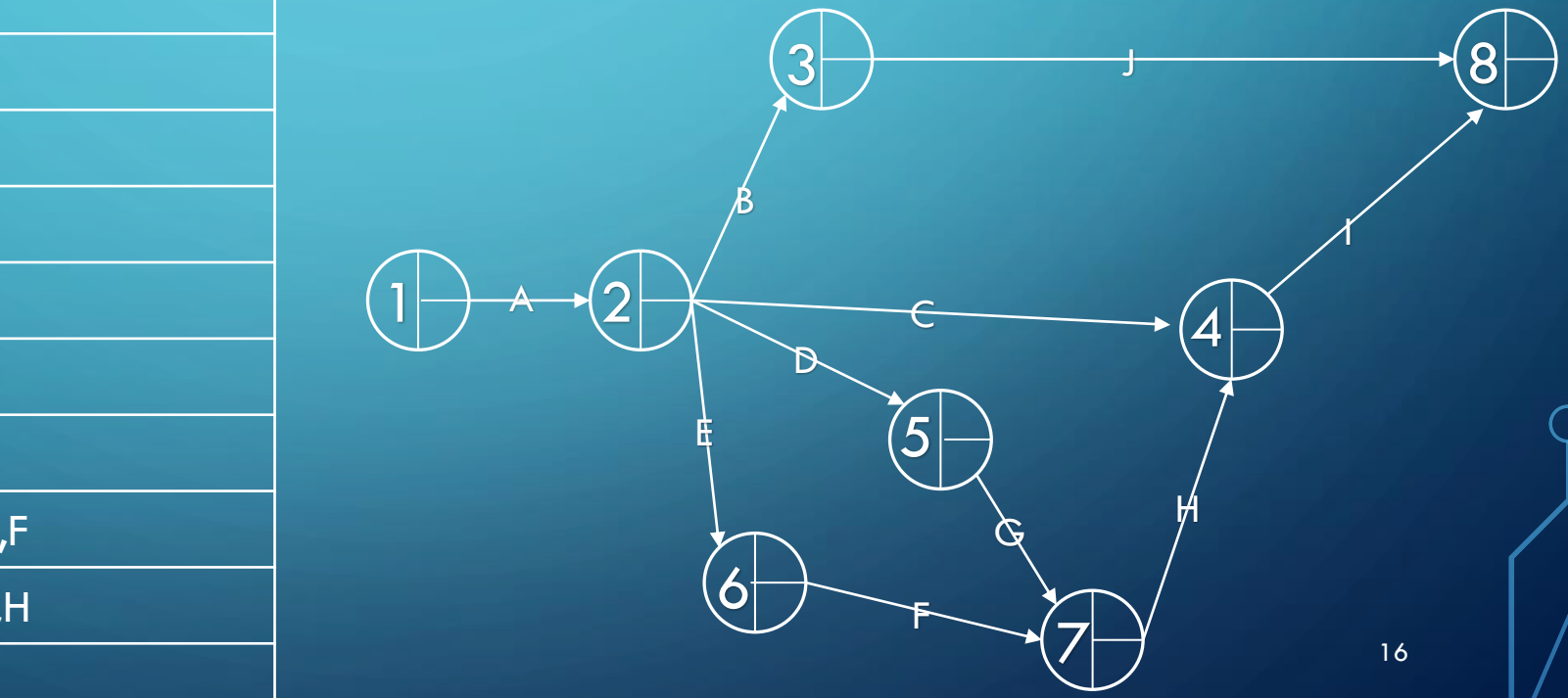


PERT CHART EXAMPLE 2

- Question

Activities	Dependencies
A	
B	A
C	A
D	A
E	A
F	E
G	D
H	G,F
I	C,H
J	B

- PERT Chart



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QUESTION

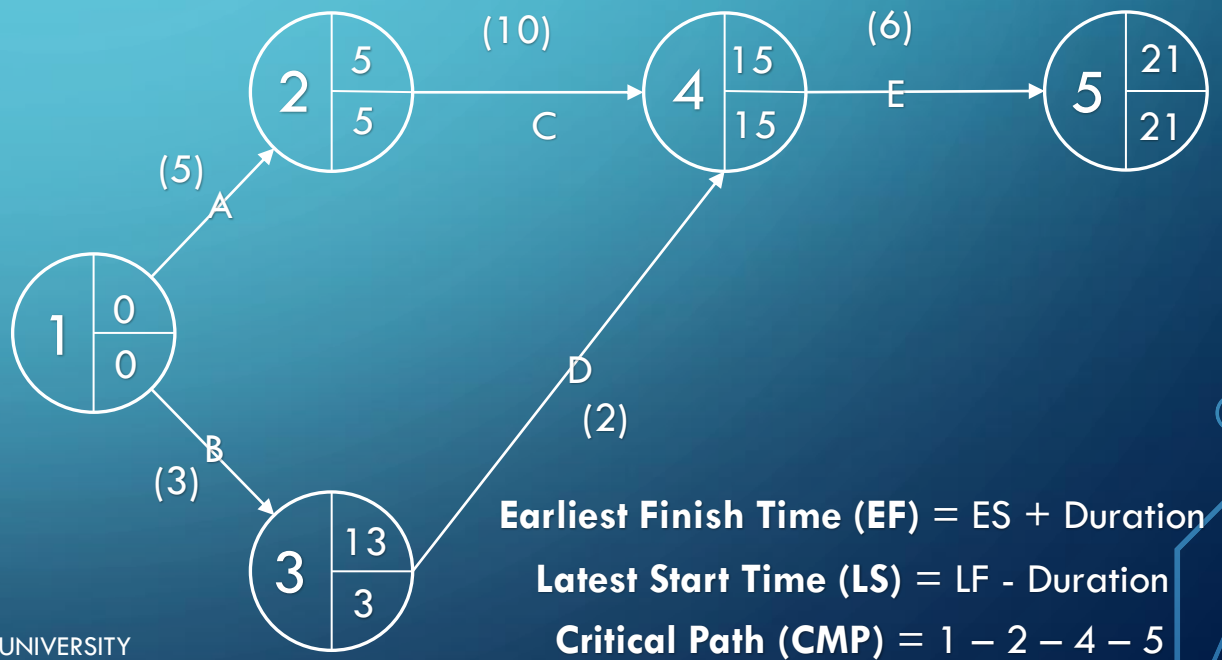
1. Construct a network diagram.
2. Find the critical path.
3. Determine the duration the project is likely to take you to complete.

PERT CHART EXAMPLE 3

1. Construct a network diagram.

Activities	Dependencies	Duration (Days)
A		5
B		3
C	A	10
D	B	2
E	C,D	6

- PERT Chart



Earliest Finish Time (EF) = ES + Duration

Latest Start Time (LS) = LF - Duration

Critical Path (CMP) = 1 – 2 – 4 – 5

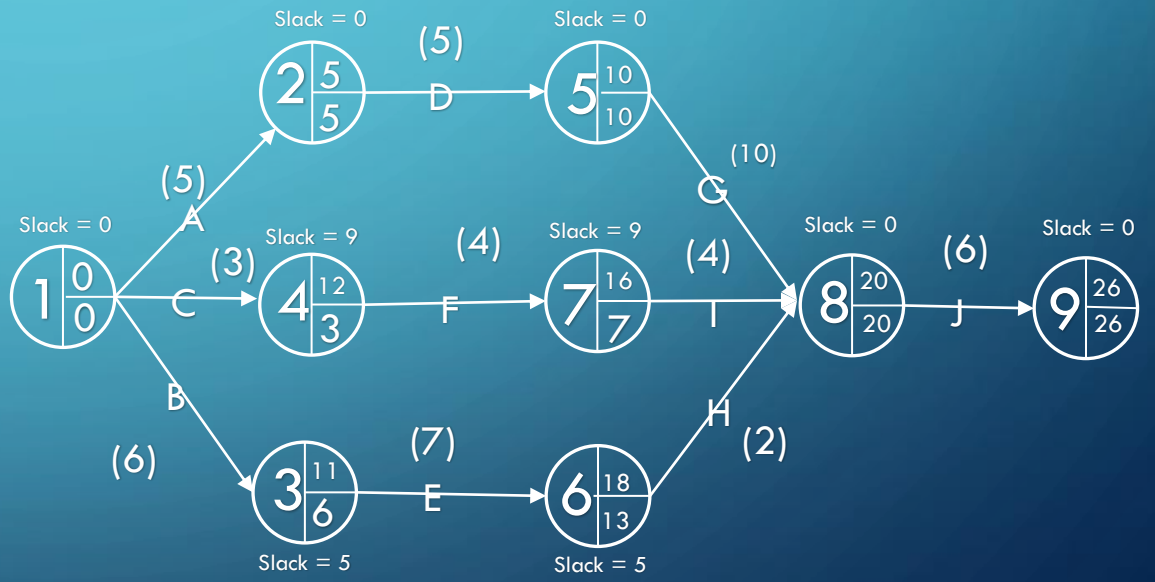
Duration of The Project = 5 + 10 + 6
= 21 Days

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PERT CHART EXAMPLE 4

Activities	Dependencies	Duration (Days)
A		5
B		6
C		3
D	A	5
E	B	7
F	C	4
G	D	10
H	E	2
I	F	4
J	G,H,I	6

- PERT Chart



Critical Path (CMP) = 1 – 2 – 5 – 8 – 9

Total Duration = 5 + 5 + 10 + 6
= 26

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PERT CHART

TASK 1

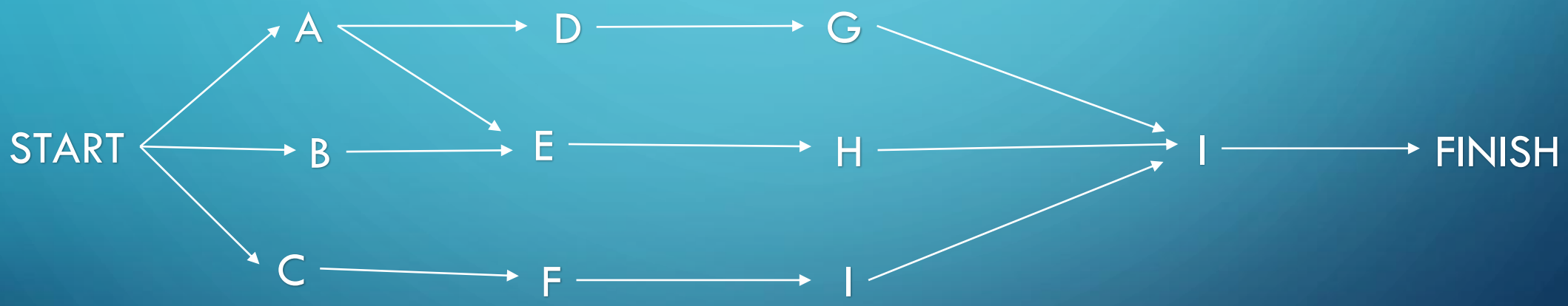
Activities	Dependencies	Duration (Days)
A		4
B		3
C		2
D	A	5
E	A,B	6
F	C	3
G	D	4
H	E	2
I	F	3
J	G,H,I	5

TASK 2

TASK ID	TASK DESCRIPTION	TASK PRE-DESSOR (S)	TASK DURATION (HOURS)
A	Project Starts		0
B	Buy Materials for A	A	10
C	Buy Materials for B	A	20
D	Build A	B,C	30
E	Build B	B,C	20
F	Finish building B	E	40
G	Join A and B	D,F	20
H	Finish Project	G	0

TASK 1

STECTCH



TASK 1 (CONTINUES…)

PERT CHART



A – D – G – J

Total Duration = 4 + 5 + 4 + 5
= 18 days

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PERT CHART GUIDE

Activity	A	B	C	D	E	F	G
Immediate Predecessors			A	A,B	D	C,E	E
Expected Time (weeks)	7	9	12	8	9	6	5

A	ES	EF
t	LS	LF

KEY

A. Activities
t. Expected duration or Activity Time
ES. Earliest Start Time
EF. Earliest Finish Time
LS. Latest Start Time
LF. Latest Finish Time

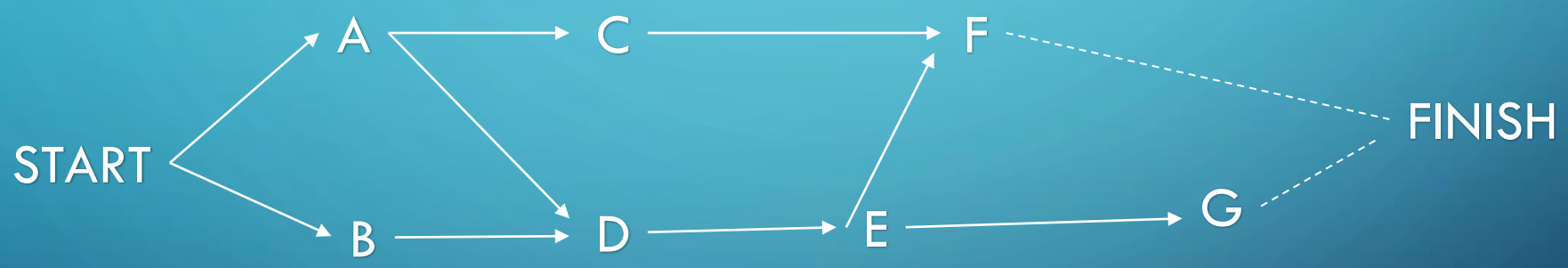
OBJECTIVES

1. Construct a project network
2. Perform forward and back passes
3. Determine project completion time
4. Calculate slack values
5. State the critical path

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STEPS WHEN CREATING A PERT CHART

1. Create a draft or sketch.

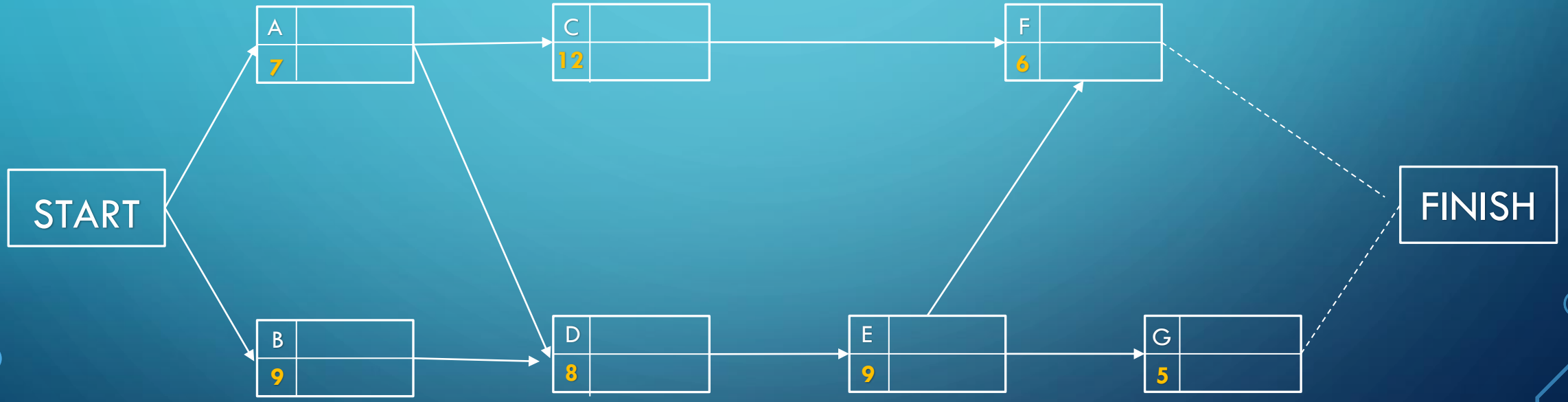


NOTE: Since F and G have nor predecessors they go to finish.

STEPS WHEN CREATING A PERT CHART (CONTINUES…)

KEY
Activity Time (t)

2. Creating the actual PERT Chart.

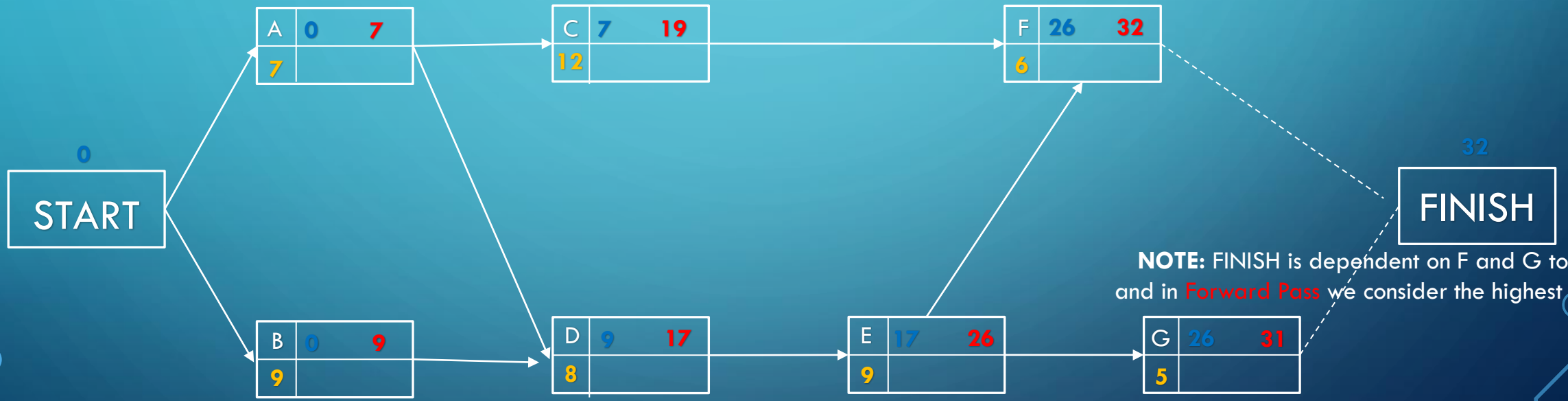


STEPS WHEN CREATING A PERT CHART (CONTINUES…)

KEY
Activity Time (t)
Forward Pass

3. Calculating the **Forward Pass: EF = ES + t**

NOTE: F is dependent on C and E to start and in Forward Pass we consider the highest EF (26)



NOTE: FINISH is dependent on F and G to stop and in Forward Pass we consider the highest EF (32)

NOTE: D is dependent on A and B to start and in Forward Pass we consider the highest EF (9)

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STEPS WHEN CREATING A PERT CHART (CONTINUES…)

KEY
Activity Time (t)
Forward Pass
Backward Pass

4. Calculating the **Backward Pass: LS = LF - t**



NOTE: A is dependent on C and D to start and in Backward Pass we consider the lowest LF (9)

NOTE: START is dependent on A and B to start and in Backward Pass we consider the lowest LF (0)

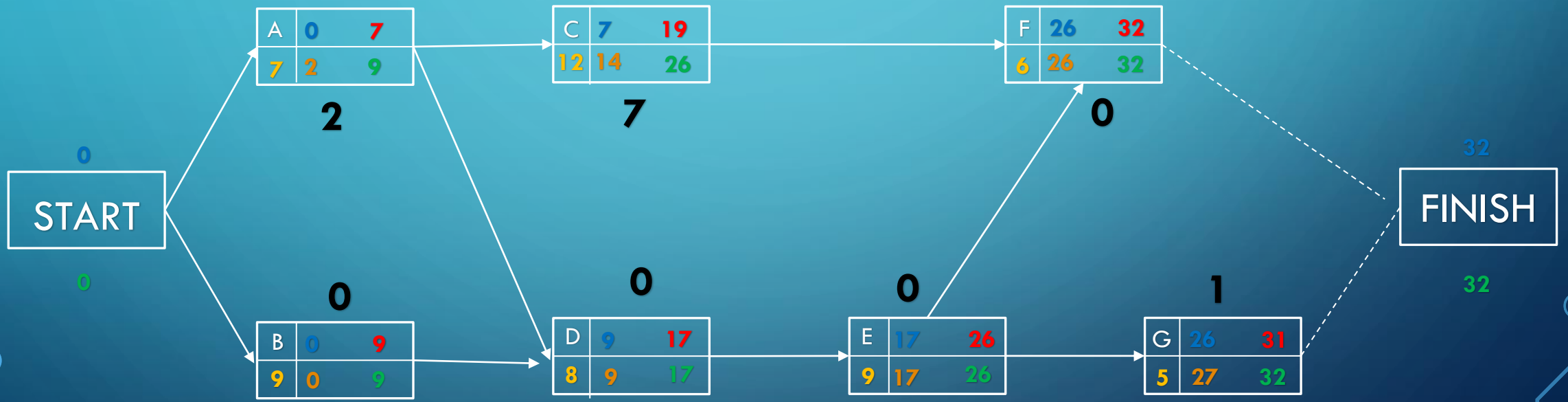
NOTE: E is dependent on F and G to start and in Backward Pass we consider the lowest LF (26)

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STEPS WHEN CREATING A PERT CHART (CONTINUES…)

KEY
Activity Time (t)
Forward Pass
Backward Pass
Slack

5. Calculating the **Slack: LS – ES or LF - EF**

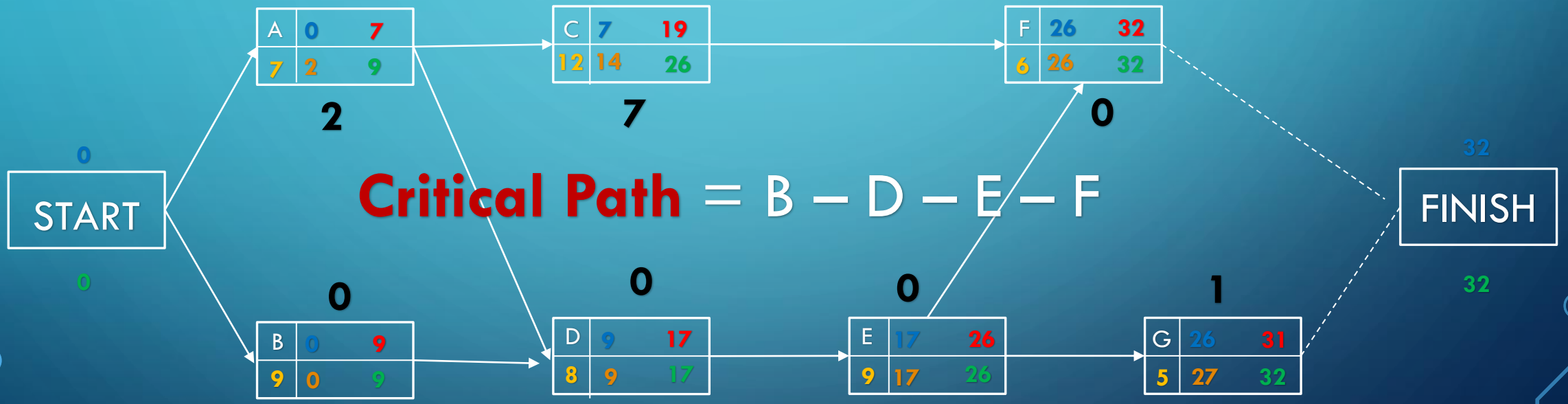


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STEPS WHEN CREATING A PERT CHART (CONTINUES…)

KEY
Activity Time (t)
Forward Pass
Backward Pass
Slack
Critical Path

6. Identifying the **Critical Path**: slack = 0



Critical Path = B – D – E – F

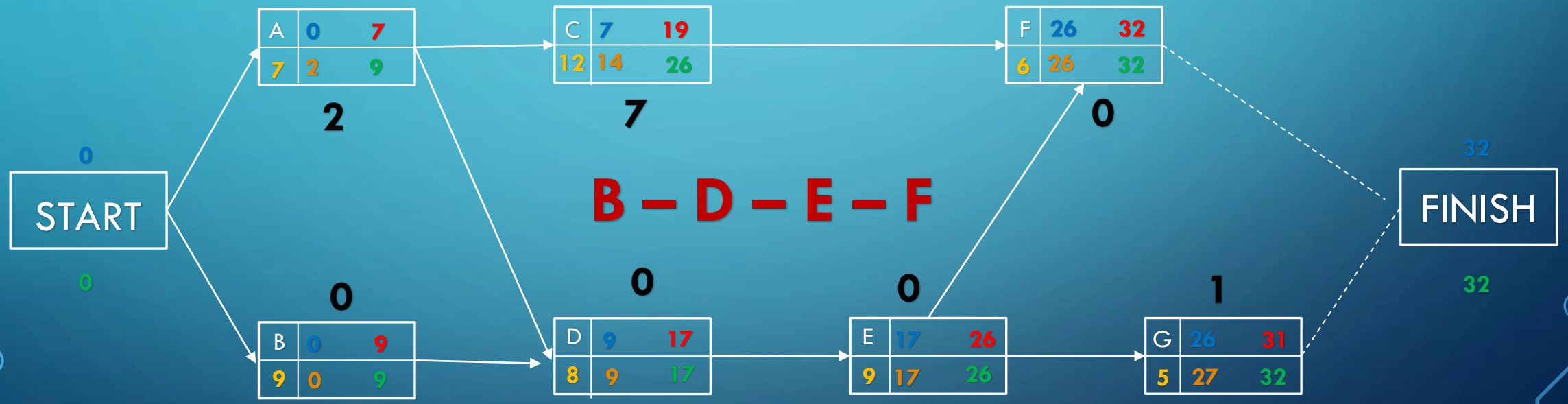
Total Duration = 9 + 8 + 9 + 6
= 32

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STEPS WHEN CREATING A PERT CHART (CONTINUES…)

KEY
Activity Time (t)
Forward Pass
Backward Pass
Slack
Critical Path

7. Final PERT Chart

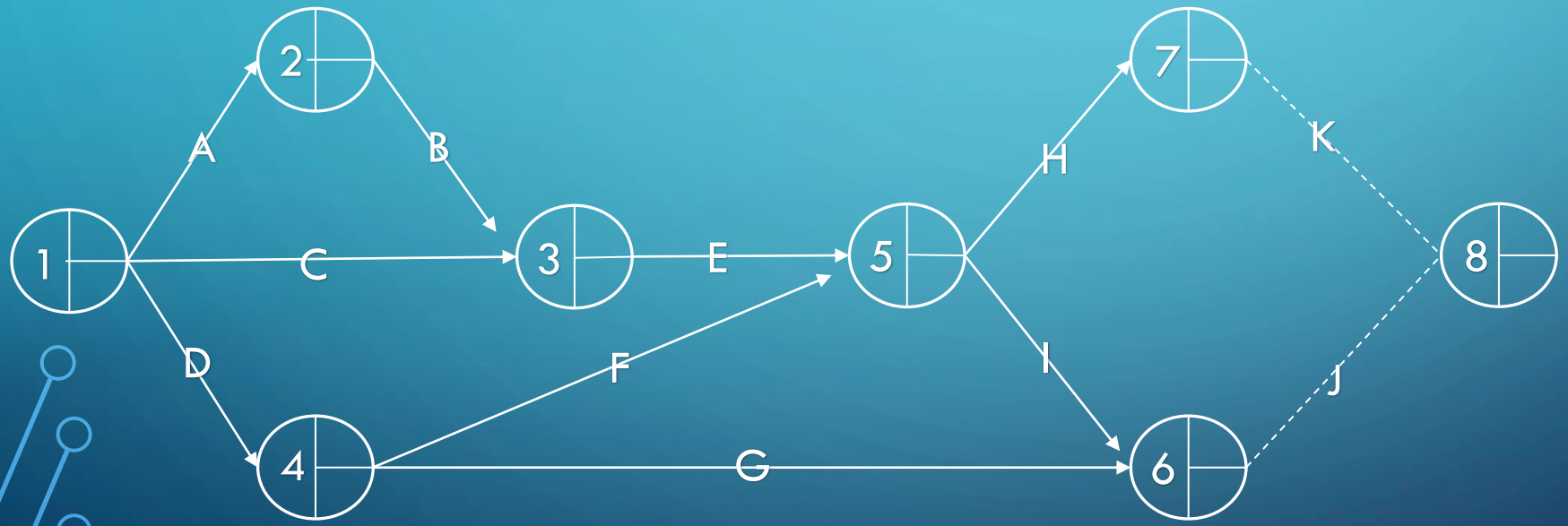


Total Duration = 9 + 8 + 9 + 6
= 32

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EXAMPLE 5

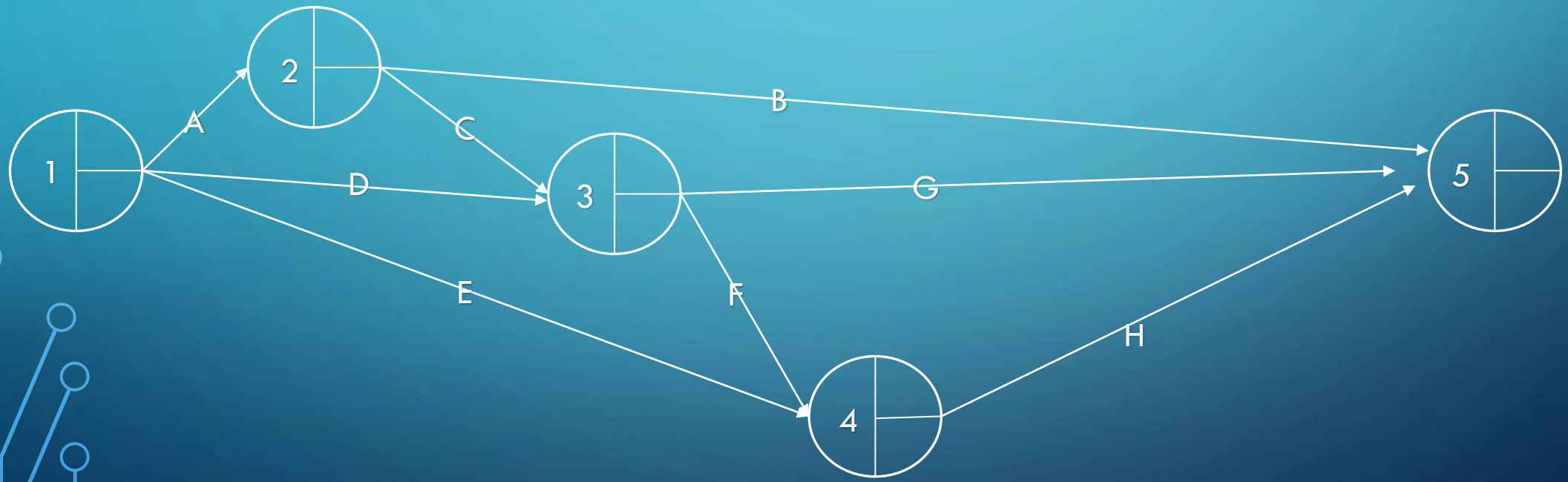
- E > B, C; F, G > D; H,I > E, F; J > I, G; K > H; B > A



Activities	Predecessors
A	
B	A
C	
D	
E	B, C
F	D
G	D
H	E, F
I	E, F
J	I, G
K	H

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EXAMPLE 6



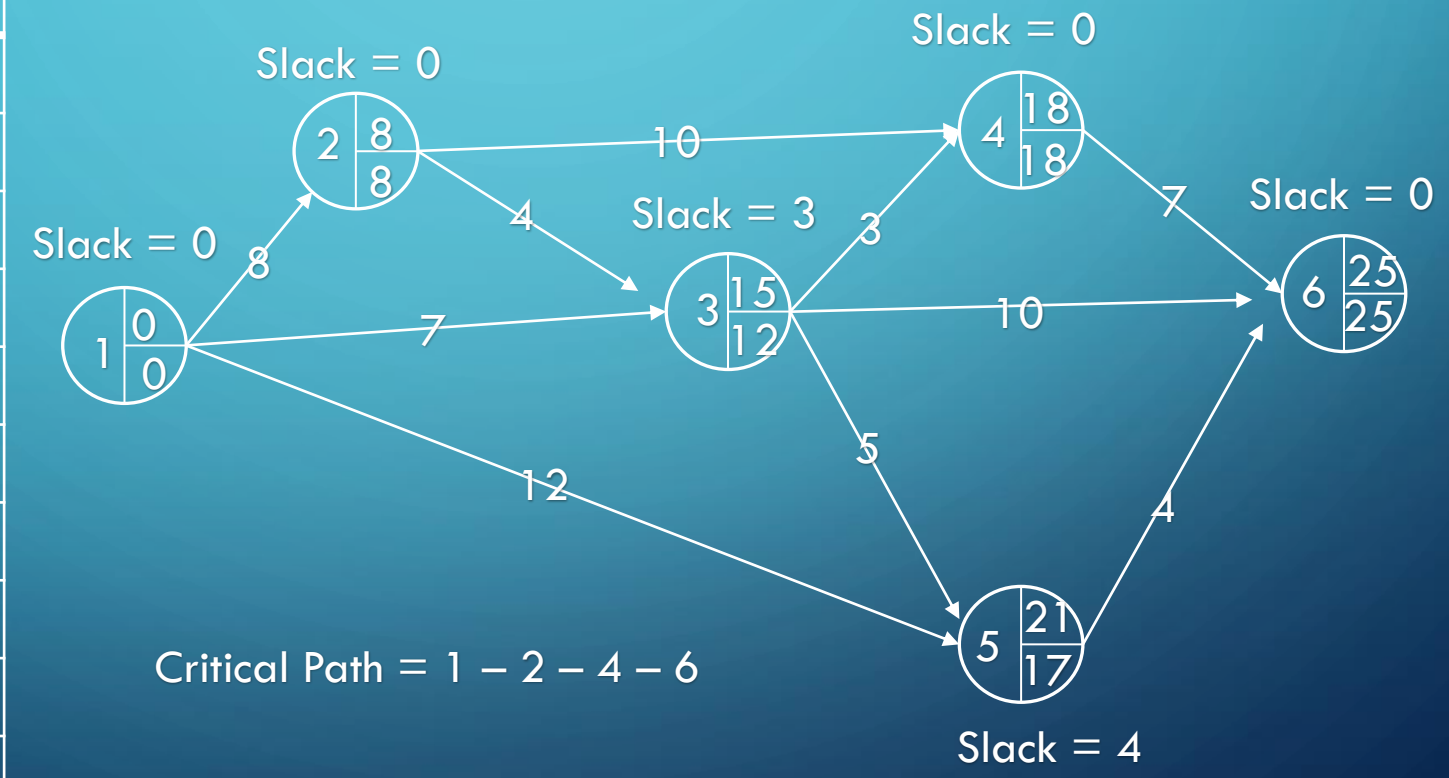
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TASK 3

1. Calculate the Earliest Finish (**EF**), Eeliest Start (**ES**), Latest Finish (**LF**) and Latest Start (**LS**).
2. Compute the Total Float (**TF**), Free Float (**FF**), Independent Float (**IND F**) and Interfering Float **(INT F**).

TASK 3 (CONTINUES…) – NETWORK DIAGRAM

Activities	Duration
1 – 2	8
1 – 3	7
1 – 5	12
2 – 3	4
2 – 4	10
3 – 4	3
3 – 5	5
3 – 6	10
4 – 6	7
5 – 6	4



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TASK 3 (CONTINUES…) – ES, EF, LF, LS & TF

Activities	ES	Duration	EF (ES + D)
1 – 2	0	8	8
1 – 3	0	7	7
1 – 5	0	12	12
2 – 3	8	4	12
2 – 4	8	10	18
3 – 4	12	3	15
3 – 5	12	5	17
3 – 6	12	10	22
4 – 6	18	7	25
5 – 6	17	4	21

Activities	LF	Duration	LS (LF - D)
1 – 2	8	8	0
1 – 3	15	7	8
1 – 5	21	12	9
2 – 3	15	4	11
2 – 4	18	10	8
3 – 4	18	3	15
3 – 5	21	5	16
3 – 6	25	10	15
4 – 6	25	7	18
5 – 6	25	4	21

LS	ES	TF (LS - ES)
0	0	0
8	0	8
9	0	9
11	8	3
8	8	0
15	12	3
16	12	4
15	12	3
18	18	0
21	17	4

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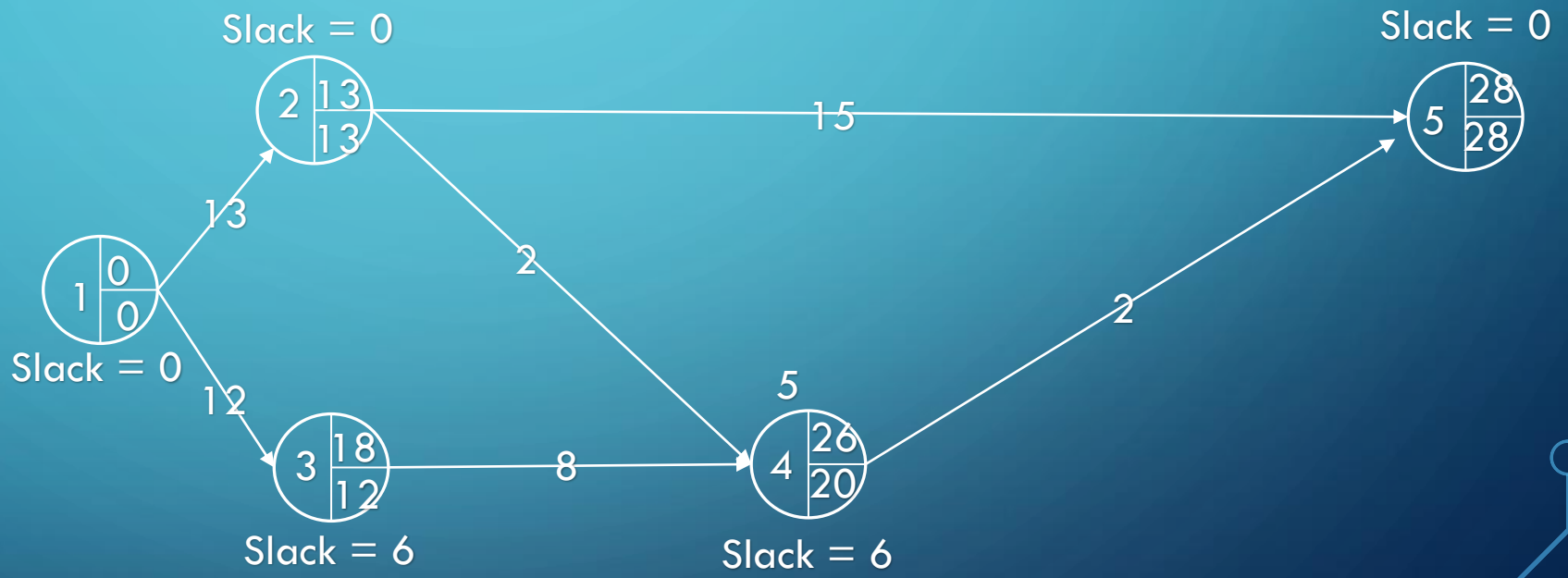
TASK 3 (CONTINUES…) – FF, IND F & INT F

Activities	T	H	TF	FF (TF – H)	IND F (FF – T)	INT F (TF – FF)
1 – 2	0	0	0	0	0	0
1 – 3	0	3	8	5	5	3
1 – 5	0	4	9	5	5	4
2 – 3	0	3	3	0	0	3
2 – 4	0	0	0	0	0	0
3 – 4	3	0	3	3	0	0
3 – 5	3	4	4	0	-3	4
3 – 6	3	0	3	3	0	0
4 – 6	0	0	0	0	0	0
5 – 6	4	0	4	4	0	0

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TASK 4

Activities	Duration
1 – 2	13
1 – 3	12
2 – 4	2
3 – 4	8
2 – 5	15
4 – 5	2



Critical Path = 1 – 2 – 5

Total Duration = 13 + 15
= 28

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TASK 4 (CONTINUES…) – ES, EF, LF, LS & TF

Activities	ES	Duration	EF (ES + D)
1 – 2	0	13	13
1 – 3	0	12	12
2 – 4	13	2	15
3 – 4	12	8	20
2 – 5	13	15	28
4 – 5	20	2	22

Activities	LF	Duration	LS (LF - D)
1 – 2	13	13	0
1 – 3	18	12	6
2 – 4	26	2	24
3 – 4	26	8	18
2 – 5	28	15	13
4 – 5	28	2	26

LS	ES	TF (LS - ES)
0	0	0
6	0	6
24	13	11
18	12	6
13	13	0
26	20	6

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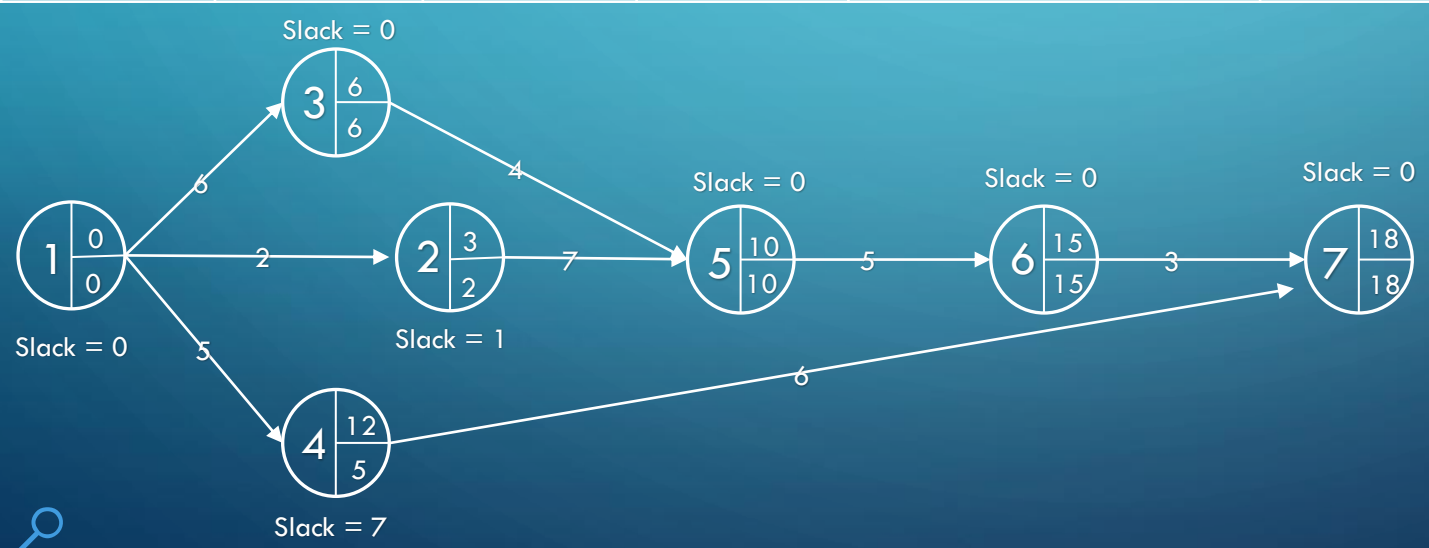
TASK 4 (CONTINUES…) – FF, IND F & INT F

Activities	T	H	TF	FF (TF – H)	IND F (FF – T)	INT F (TF – FF)
1 – 2	0	0	0	0	0	0
1 – 3	0	6	6	0	0	6
2 – 4	0	6	11	5	5	6
3 – 4	6	6	6	0	-6	6
2 – 5	0	0	0	0	0	0
4 – 5	6	0	6	6	0	0

UP NEXT… EXAMPLE 7

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Activity	Optimistic (O or to)	Pessimistic (P or tp)	Most Likely (M or tm)	4tm	Te = (to + 4tm + tp) / 6	σ = (tp – to) / 6	σ²
1 – 2	1	3	2	4(2) =8	(1 + 8 + 3) / 6 = 2	(3 – 1) / 6 = 0.33	0.10
1 – 3	5	7	6	4(6) = 24	(5 + 12 + 7) / 6 = 6	(7 – 5) / 6 = 0.33	0.10
1 – 4	3	7	5	4(5) = 20	(3 + 20 + 7) / 6 = 5	(7 – 3) / 6 = 0.66	0.43
2 – 5	5	9	7	4(7) = 28	(5 + 28 + 9) / 6 = 7	(9 – 5) / 6 = 0.66	0.43
3 – 5	2	6	4	4(4) = 16	(2 + 16 + 6) / 6 = 4	(6 – 2) / 6 = 0.66	0.43
5 – 6	4	6	5	4(5) = 20	(4 + 20 + 6) / 6 = 5	(6 – 4) / 6 = 0.33	0.10
4 – 7	4	8	6	4(6) = 24	(4 + 24 + 8) / 6 = 6	(8 – 4) / 6 = 0.66	0.43
6 – 7	2	4	3	4(3) = 12	(2 + 12 + 4) /6 = 3	(4 – 2) / 6 = 0.33	0.10



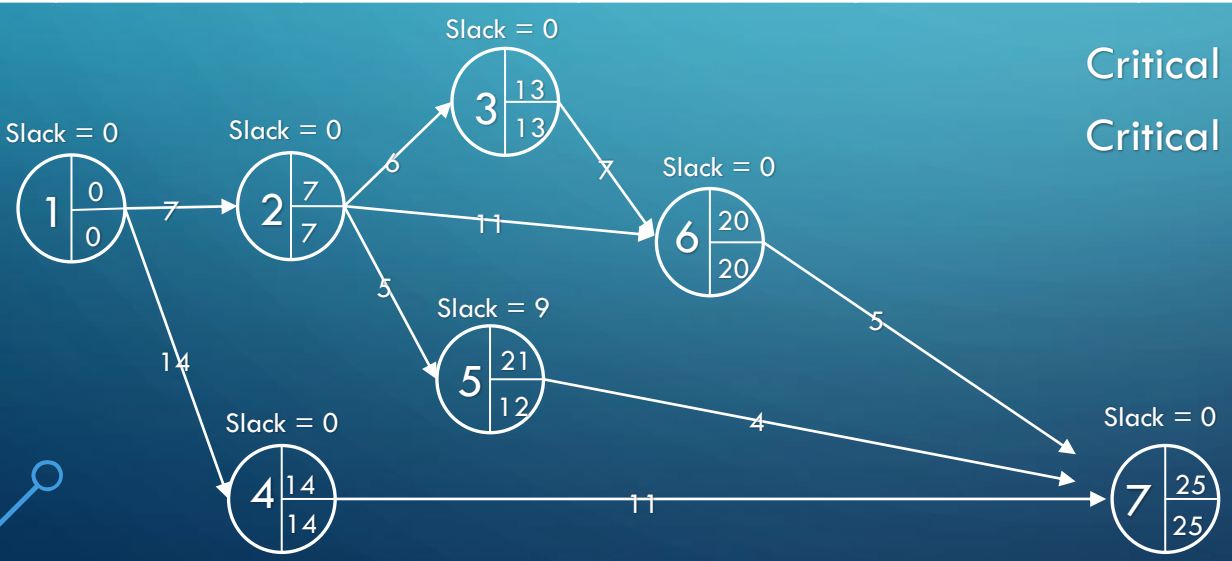
Critical Path = 1 – 3 – 5 – 6 – 7

Critical Variance = 0.10 + 0.43 + 0.10 + 0.10
= 0.73

Total Duration = 6 + 4 + 5 + 3
= 18

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Activity	Optimistic (O or to)	Pessimistic (P or tp)	Most Likely (M or tm)	4tm	Te = to + 4tm + tp	σ = (tp – to) / 6	σ²
1 – 2	3	15	6	4(6) = 24	(3 + 24 + 15) / 6 = 7	(15 – 3) / 6 = 2	4
2 – 3	2	14	5	4(5) = 20	(2 + 20 + 14) / 6 = 6	(14 – 2) / 6 = 2	4
1 – 4	6	30	12	4(12) = 48	(6 + 48 + 30) / 6 = 14	(30 – 6) / 6 = 4	16
2 – 5	2	8	5	4(5) = 20	(2 + 20 + 8) / 6 = 5	(8 – 2) / 6 = 1	1
2 – 6	5	17	11	4(11) = 44	(5 + 44 + 17) / 6 = 11	(17 – 5) / 6 = 2	4
3 – 6	3	15	6	4(6) = 24	(3 + 24 + 15) / 6 = 7	(15 – 3) / 6 = 2	4
4 – 7	3	27	9	4(9) = 36	(3 + 36 + 27) / 6 = 11	(27 – 3) / 6 = 4	16
5 – 7	1	7	4	4(4) = 16	(1 + 16 + 7) / 6 = 4	(7 – 1) / 6 = 1	1
6 – 7	2	8	5	4(5) = 20	(2 + 20 + 8) / 6 = 5	(8 – 2) / 6 = 1	1



Critical Path = 1 – 4 – 7

Critical Variance = 16 + 16
= 32

Total Duration = 14 + 11
= 25

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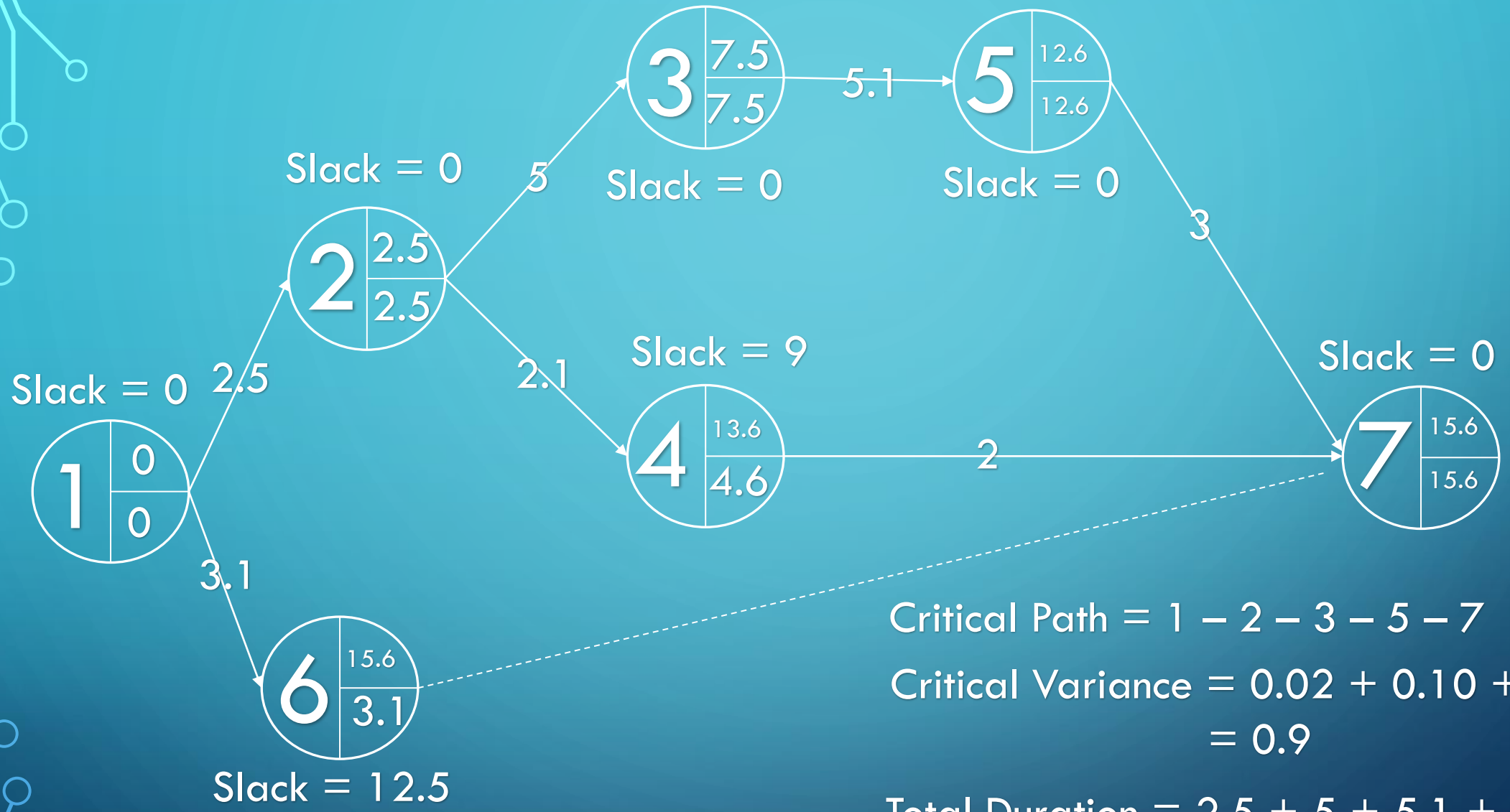
TASK 5

Activity	O	P	M	Dependences
Identify compromised systems	1	2	3	None
Patch vulnerabilities	2	4	6	Identify compromised vulnerabilities
Secure User Accounts	1	4	2	Patch vulnerabilities
Restore data	3	8	5	Patch vulnerabilities
Implement additional controls	2	5	3	Secure User Accounts
Revise Policies	1	3	2	Identify compromised systems
Final audit and testing	2	4	3	Implement additional controls, restore data

TASK 5 (CONTINUES…)

		Dependence		Activity	to	tp	tm	4tm	Te	σ	σ^2
1	Identify Compromised Systems										
2	Path Vulnerability	1		1 – 2	1	2	3	12	2.5	0.16	0.02
3	Secure User Accounts	2		2 – 3	2	4	6	24	5	0.33	0.10
4	Restore Data	2		2 – 4	1	4	2	8	2.1	0.5	0.25
5	Implement additional Controls	3		3 – 5	3	8	5	20	5.1	0.83	0.68
6	Revise polices	1		1 – 6	2	5	3	12	3.1	0.5	0.25
7	Final Audit and Testing	5,4		4 – 7	1	3	2	8	2	0.33	0.10
				5 – 7	2	4	3	12	3	0.33	0.10

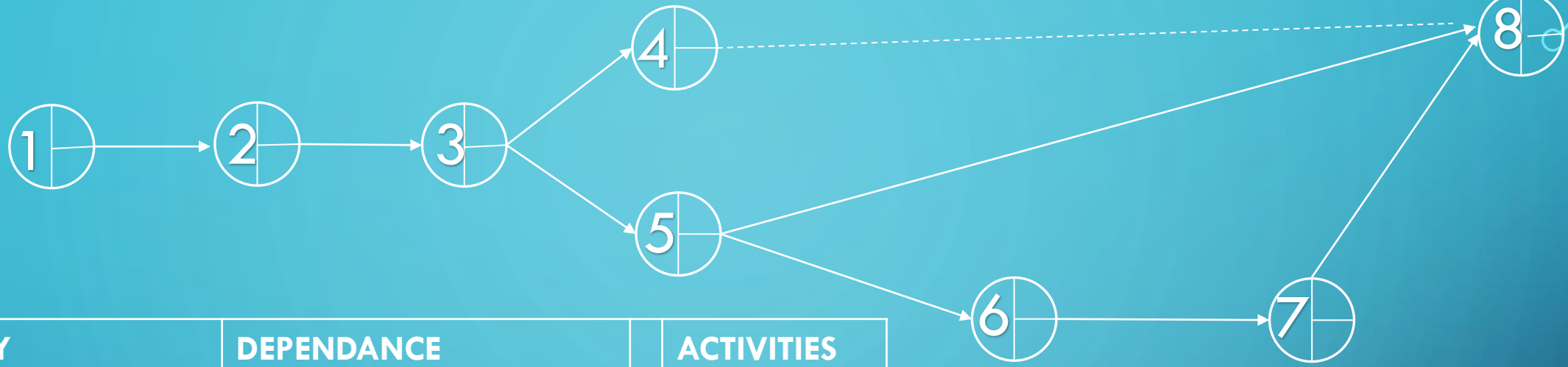
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Critical Path = 1 – 2 – 3 – 5 – 7

Critical Variance = 0.02 + 0.10 + 0.68 0.10
= 0.9

Total Duration = 2.5 + 5 + 5.1 + 3
= 15.6



	ACTIVITY	DEPENDANCE		ACTIVITIES
1	Root cause analysis			**1 – 2**
2	Patch development	Root cause analysis		**2 – 3**
3	Pact testing	Patch development		**3 – 5**
4	Schedule Maintenance	Pact testing		**3 – 4**
5	Deploy patch	Pact testing		*4 – 5*
6	Validate remediation	Deploy patch		**5 – 6**
7	Update documentation	Validate remediation		**6 – 7**
8	Notify stakeholders	Update documentation, Deploy patch		*7 – 8* *5 – 8*

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